Signal Processing in Practice (E9:222)

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Linear Prediction Problem

- Consider a WSS $x[n]$
- The predicted output is given by $\hat{x}_N^f[n] = -\sum_{i=1}^{N} a_{N,i}^* x[n-i]$
- The error

$$e_N^f[n] = x[n] - \hat{x}_N^f[n] = x[n] + \sum_{i=1}^{N} a_{N,i}^* x[n-i]$$

- The coefficients $\{a_{N,i}\}_{i=1}^{N}$ are given by min $E\{|e_N^f[n]|^2\}$
- Filtering analogy

$$x[n] \rightarrow \boxed{A_N(z)} \rightarrow e_N^f[n] \qquad e_N^f[n] \rightarrow \boxed{\frac{1}{A_N(z)}} \rightarrow x[n]$$

Adobe Acr

Solving Linear Prediction Problem

- Orthogonality Principle - optimal LP model satisfies
  - Error signal $e_N^f[n] \perp \{x[n-i]\}_{i=1}^N$ which means

$$E\{e_N^f[n]x^*[n-i]\} = 0$$

- Using the above set of equations $\boldsymbol{Ra} = \boldsymbol{r}$

$$\boldsymbol{R} = \begin{bmatrix} R(0) & R(1) & \dots & R(N-1) \\ R^*(1) & R(0) & \dots & R(N-2) \\ \dots & \dots & \dots & \dots \\ R^*(N-1) & R^*(N-2) & \dots & R(0) \end{bmatrix} \quad \boldsymbol{a} = \begin{bmatrix} a_{N,1} \\ a_{N,2} \\ \dots \\ a_{N,N} \end{bmatrix} \quad \boldsymbol{r} = -\begin{bmatrix} R^*(1) \\ R^*(2) \\ \dots \\ R^*(N) \end{bmatrix}$$

- Normal/Yule-walker/Weiner-Hopf Equations

Properties

- Minimum phase property for $A_N(z)$

- Autocorrelation matrix is Hermitian positive semi-definite, Toeplitz

- Min. err $\epsilon_N^f = E\{|e_N^f[n]|^2\} = R(0) + \sum_{i=1}^{N} a_{N,i}^* R^*(i)$

- Using the above expression, we get augmented normal equations.

$$\boldsymbol{R}_{N+1}\boldsymbol{a}_{N+1} = \begin{bmatrix} \epsilon_N^f \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

- One-to-one mapping between autocorrelation sequence and the linear prediction coefficients and the min. error.

Properties

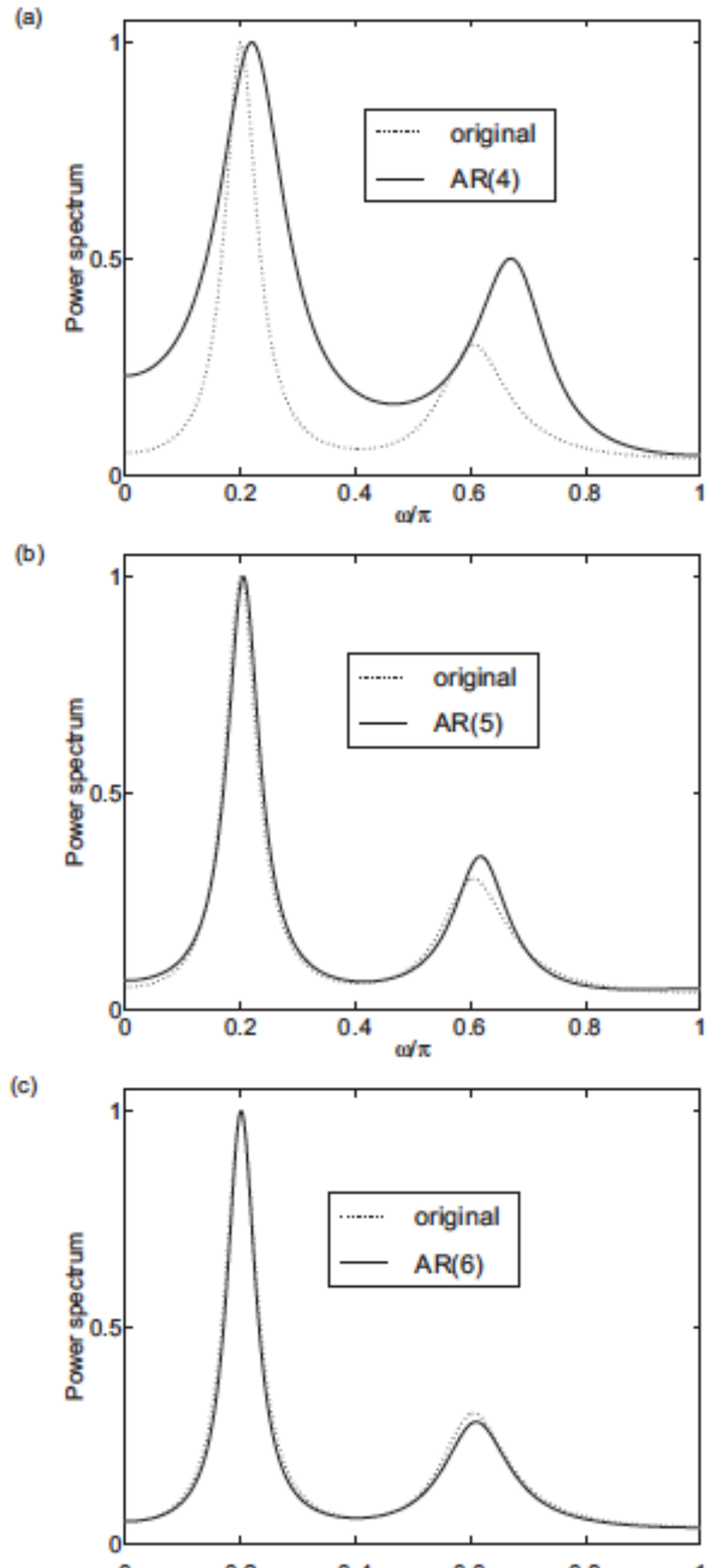
- Output of filter with $V(z) = v_0^* + v_1^* z^{-1} + \ldots + v_{N-1}^* z^{-(N-1)}$

$$\boldsymbol{x}[n] = \begin{bmatrix} x[n] \\ x[n-1] \\ \ldots \\ x[n-N+1] \end{bmatrix} \qquad \boldsymbol{v} = \begin{bmatrix} v_0 \\ v_1 \\ \ldots \\ v_{N-1} \end{bmatrix} \qquad y[n] = \boldsymbol{v}^H \boldsymbol{x}[n]$$

- If $\boldsymbol{v}$ represents a unit eigenvector of autocorrelation matrix $\boldsymbol{R}_N$

$$E[|y[n]|^2] = \lambda$$

- Relationship between eigenvalues and power spectrum of the signal.


(a)
original
AR(4)
Power spectrum
ω/π
(b)
original
AR(5)
Power spectrum
ω/π
(c)
original
AR(6)
Power spectrum

Questions to solve

5. **Implementing Linear Prediction** - Use the signals provided in the following data set,

   http : *//leap.ee.iisc.ac.in/sriram/teaching/SIP/HW3DataFiles.tar.gz*

   (a) Apply a linear prediction of order N on each short-term Fourier transform window (25 ms with a shift of 10ms) with a Hamming window. Plot the energy of residual signal for each STFT frame for a given order N. Can you relate the energy of the residual signal with the underlying speech sound in the given frame.

   (b) Reconstruct the signal using AR model of the LP spectrum and the original STFT phase. What happens if the inverse LP filter formulation is used with (a) white noise excitation on all STFT frames, (b) pitch based impulse train on all voiced frames and white noise on unvoiced frames. How does the quality of reconstruction vary with N. Does it saturate after a small choice of N.

   (Points 35)

HW3DataFiles.tar.gz

6. **Spectral Transform Linear Prediction** - A transform version of the autocorrelation sequence can be computed using inverse DTFT of $S_{xx}(e^{j\omega})^{\frac{1}{q}}$, where q is a positive integer. Using the autocorrelation sequence one can derive LP coefficients and the AR model of the spectrum $\hat{S}_{xx}(e^{j\omega}$ which approximates $S_{xx}(e^{j\omega})^{\frac{1}{q}}$. If we compute $\hat{S}_{xx}(e^{j\omega})^q$, it approximates the original PSD $S_{xx}(e^{j\omega}$. For example, $q = 1$ is the standard LP problem. For a speech signal frame (from the signal above), implement STLP and plot $S_{xx}(e^{j\omega}$ and $\hat{S}_{xx}(e^{j\omega})^q$ for $q = 1, 2, 3$. Comment on the approximation properties for different choices of q and for a voiced and unvoiced frame. **(Points 15)**